MGMT-6216 · PREDICTIVE ANALYTICS · FANSHAWE COLLEGE

Clustering Socio-Economic Profiles of Countries

K-Means (PAM), hierarchical clustering and PCA in R

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Executive summary

2 clusters

clear groups: 69 less-developed and 98 more-developed countries.

87%

agreement between K-Means and hierarchical clustering

64%

of variance in two principal components. PCA confirms the split is genuine.

What we found

Both methods split 167 countries into the same two profiles. The less-developed group has lower income and GDP per capita, higher child mortality and fertility, and lower life expectancy. The more-developed group shows the opposite pattern.

Which clustering technique is more suitable?

K-Means (PAM) is the most interpretable and stable method for this dataset, and the other two checks support it.

Criteria for grouping the countries

8 economic and health indicators for 167 countries.

Economic

Income
net income per person

Exports
trade out, as a share of the economy

Imports
trade in, as a share of the economy

Inflation
how fast prices are rising

Health

Child mortality
deaths of children under 5 per 1,000 births

Life expectancy
average years a newborn is expected to live

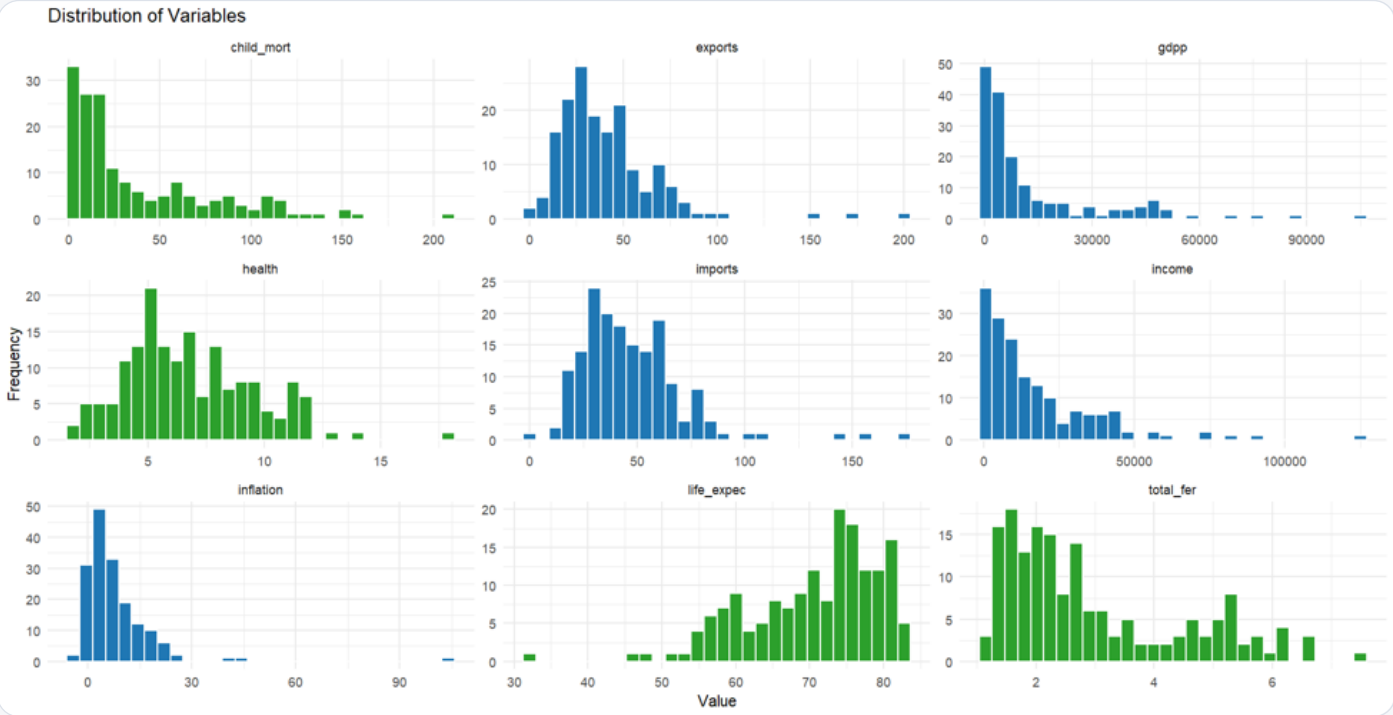
Fertility rate
average number of children per woman

Health spending
what the country spends on health

Left out: GDP per capita, because it carries almost the same signal as income (correlation 0.90).

Several indicators are heavily skewed

The data reflects inequality among countries: only a small number are rich.



Financial (blue)

GDP per capita, income, inflation: very strong right skew with many outliers.

Imports, exports: right-skewed, but more evenly spread with fewer outliers.

Health (green)

Child mortality, fertility rate: right-skewed. **Life expectancy:** left-skewed.

Health spending: the most symmetric, with fewer outliers.

Preparing the data for distance-based clustering

1 · Check

No missing values and no duplicate rows across the 167 countries.

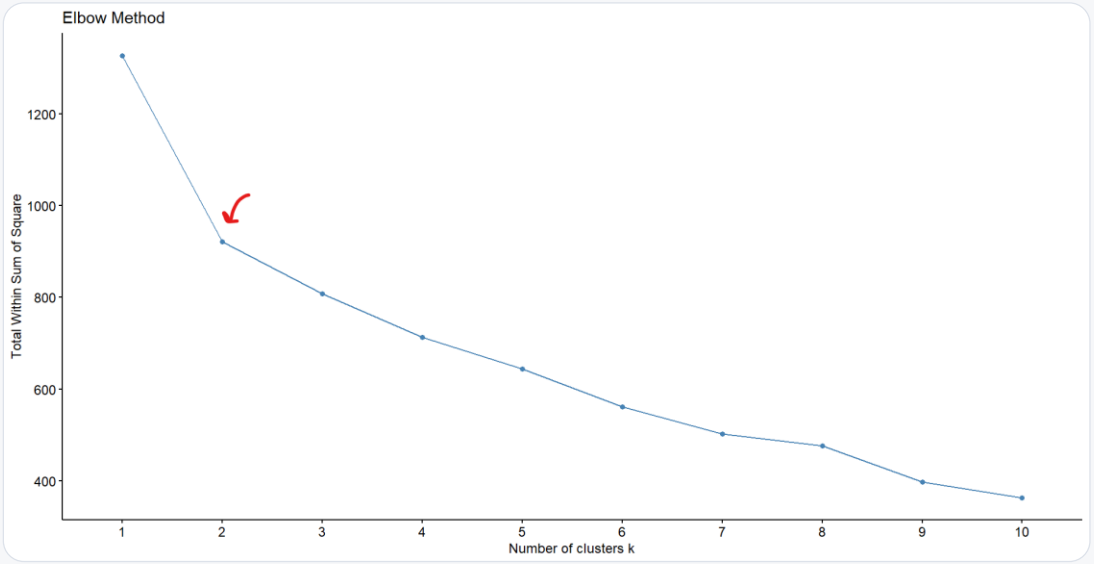
2 · Select

Country name and GDP per capita left out of the clustering input, keeping 8 indicators.

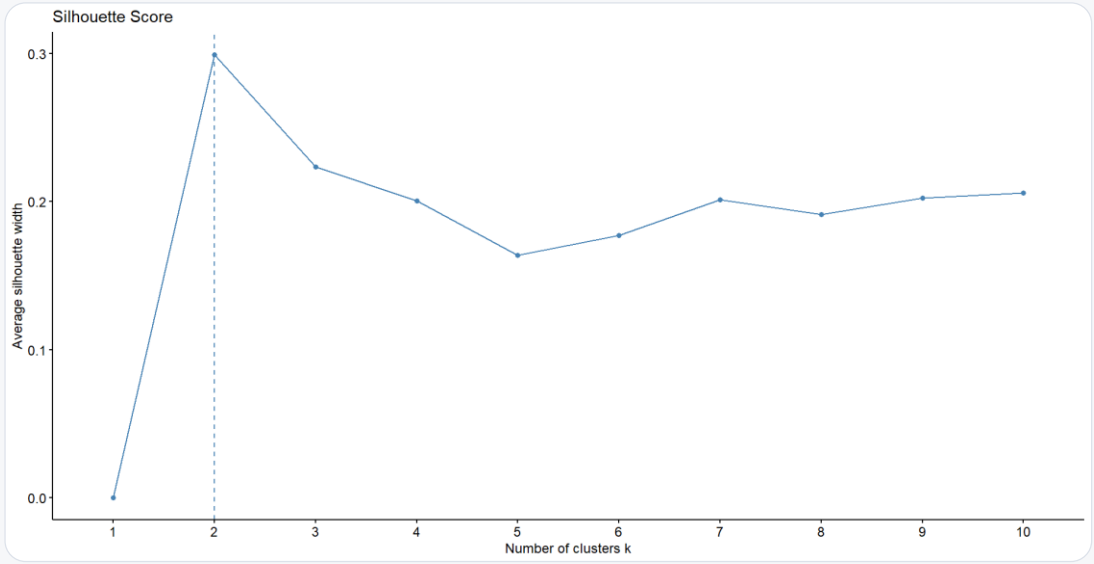
3 · Scale

Z-score scaling, so no indicator dominates the distances because of its units.

Elbow and silhouette both point to two clusters

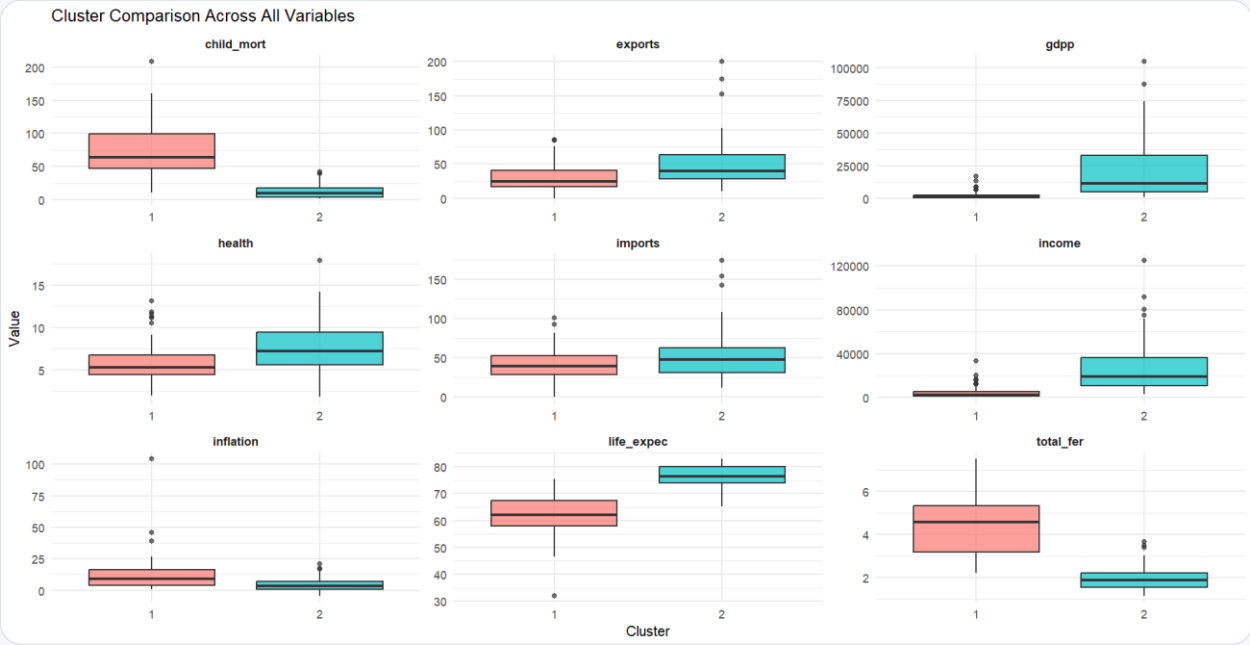


Elbow: the curve bends at k = 2.



Silhouette: the best average width (0.3) is at k = 2.

Characteristics of the 2 clusters



Median	Cluster 1 69 countries	Cluster 2 98 countries
Name	Less-developed	More-developed
Child mortality	64.4	10.3
Life expectancy	62.2	76.4
Fertility rate	4.56	1.88
Income	2,660	19,250
GDP per capita	1,170	11,600

Table shows medians.

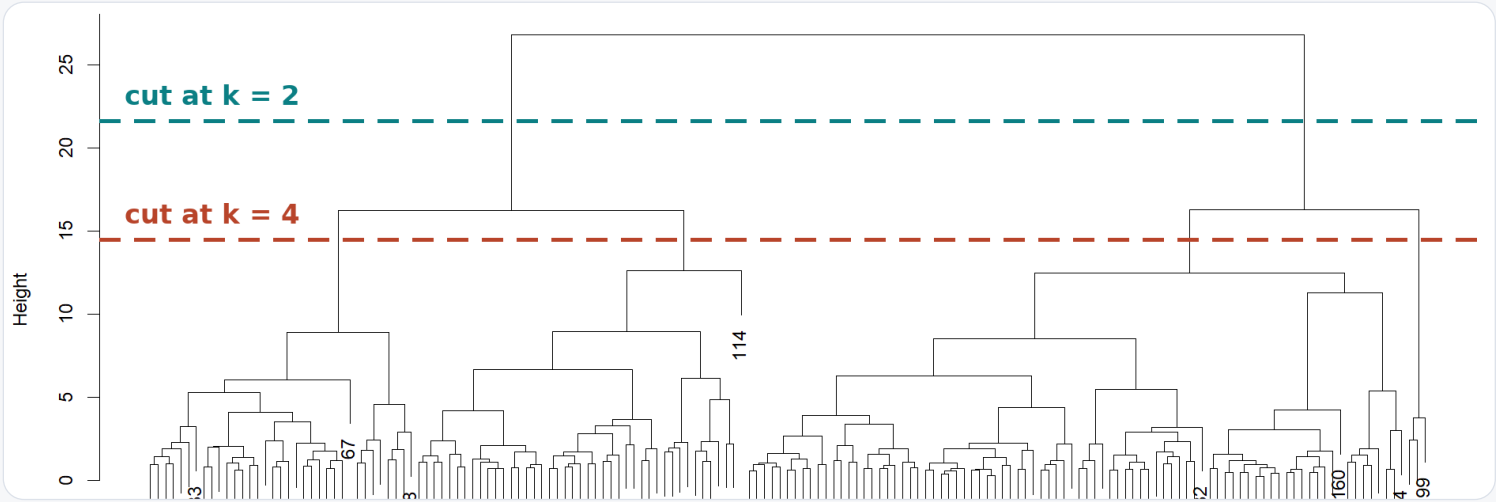
Cluster 1 · less-developed

Higher child mortality, fertility and inflation.
Lower life expectancy, income and GDP per capita.

Cluster 2 · more-developed

The reverse, plus higher health spending, exports and imports.

Hierarchical clustering gives a second opinion



Dendrogram (AGNES, Ward linkage): each leaf is a country, and the height shows how different the groups are when they merge.

0.95

agglomerative coefficient:
a strong clustering structure

87%

same group as K-Means at k = 2
(146 of 167 countries)

Why not more clusters? Testing k = 4 gave weaker groups

0.20

average silhouette, down from about 0.3. One cluster scores only 0.12.

67%

agreement between K-Means and hierarchical, vs 87% at k = 2.

3

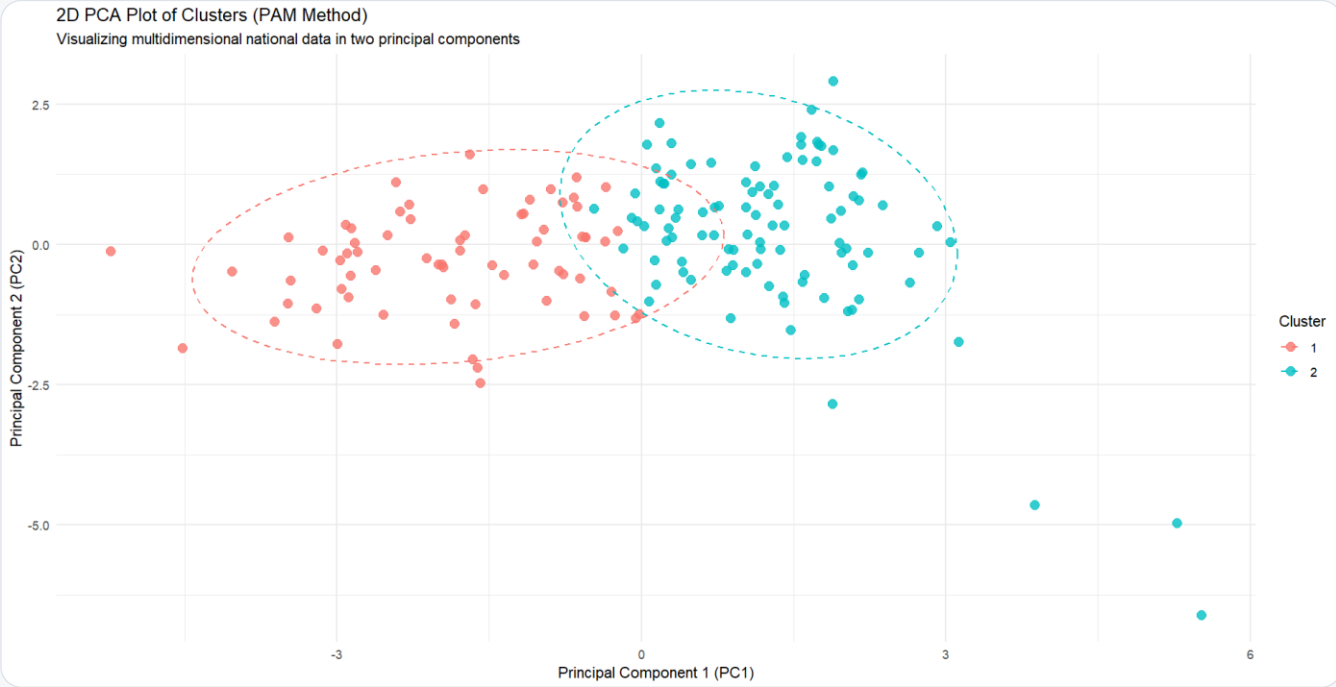
countries in one hierarchical cluster: an outlier group, not a real segment.

PCA and comparison table confirm the 2 clusters

64%

of the total variance is captured by the first two components (PC1 44.7%, PC2 19.3%).

The clusters separate along PC1, with only a small overlap in the middle.



K-Means vs hierarchical clustering

	Hierarchical 1	Hierarchical 2
K-Means 1	63	6
K-Means 2	15	83

146 of 167 countries (87%) land in the same group under both methods.

What the analysis shows

Two profiles

A less-developed group of 69 countries and a more-developed group of 98, separated on income, health and fertility.

K-Means (PAM) fits best

Clear, easy-to-interpret groups, less affected by early grouping decisions than hierarchical clustering. PCA validates the separation.

Skills shown

Data checks and EDA, scaling, choosing k, K-Means and hierarchical clustering, PCA, and explaining the results plainly.